

A low-diversity ordinal regime produces the excess3 peak near the logistic accumulation point

Johel Padilla-Villanueva

Department of Environmental Health

University of Puerto Rico Medical Sciences Campus

ORCID: 0000-0002-5797-6931

johelpadilla@gmail.com

24 September 2026

DOI: 10.5281/zenodo.22945800

Abstract

On four diffusively coupled logistic maps, the hybrid proxy $\text{excess3} = 0.6 \text{Syn} + 0.4 \text{Surp}$ reaches its maximum near the period-doubling accumulation and is lower in developed chaos. The geometric reading of that maximum was left as a hypothesis. The same generator, seeds, and hybrid are decomposed here on a grid of 49 parameter values, with six realizations at each value. The six-seed mean is largest at $r = 3.5675$, where $\text{excess3} = 2.378 \pm 0.002$ in windows of 13 symbols. Surprise against independence carries that elevation. The synergy term is 0.0065 ± 0.0004 there and attains its own maximum only at $r = 3.95$. Inside the short window the joint occupies 4.11 ± 0.02 ordinal tuples. Independent draws from the pairwise maximum-entropy projection of the invariant joint reproduce the score of independent draws from the joint itself. The sliding window lies about 0.30 bits higher because consecutive symbols are more concentrated than independent samples. The Kullback–Leibler residual beyond pairwise margins is 0.0003 ± 0.0001 on the full symbol block at the maximum and 0.192 ± 0.002 at $r = 3.85$, where the proxy has already fallen. A period-3 window does not repeat the peak. Lengthening the window from 6 to 312 symbols leaves the accumulation score near 2.3 and lowers the chaotic score from about 2.08 to 0.51. The high values form a band, $r \in [3.544, 3.595]$, rather than a cusp at the accumulation point. On this generator the proxy and the irreducible residual mark different parts of the cascade.

Keywords: excess3; Bandt–Pompe patterns; Feigenbaum accumulation; pairwise residual; coupled logistic maps

Resumen. En cuatro mapas logísticos acoplados, el proxy $\text{excess3} = 0.6 \text{Syn} + 0.4 \text{Surp}$ alcanza su máximo cerca de la acumulación de doblamiento de periodo. En una rejilla de 49 valores de r , con seis realizaciones, ese máximo está en $r = 3.5675$ y vale 2.378 ± 0.002 para ventanas de 13 símbolos. La elevación la lleva la sorpresa frente a la independencia. La sinergia vale ahí 0.0065 ± 0.0004 y solo crece más adelante. La ventana ocupa 4.11 ± 0.02 tuplas ordinales conjuntas. La proyección de máxima entropía por pares reproduce el valor de extracciones independientes de la ley conjunta; la ventana deslizante queda unos 0.30 bits por encima porque los símbolos consecutivos están más concentrados. El residuo de Kullback–Leibler más allá de los pares es 0.0003 ± 0.0001 en el bloque completo del máximo y

0.192 ± 0.002 en $r = 3.85$, donde el proxy ya bajó. La ventana de periodo 3 no repite el pico. Alargar la ventana mantiene el pico y agranda el contraste con el caos. En este generador el proxy y el residuo irreducible marcan tramos distintos de la cascada.

1 Introduction

A critical transition is often described through the growth of variance and lag-1 autocorrelation [Scheffer et al., 2009]. An ordinal reading asks a different question: whether the joint order of several series is being reorganized [Bandt and Pompe, 2002, Padilla-Villanueva, 2026b]. In that reading, Systemic Tau summarizes pairwise rank concordance, and a discrete clock is built from nested ordinal conjunctions. The third level of the clock is the residual of the joint ordinal law beyond pairwise margins. The hybrid `excess3` was specified so that a short window could be scored when that residual is costly to compute [Padilla-Villanueva, 2026a,c]. The weights 0.6 and 0.4 are a declared design choice. They were not fitted to the logistic suite.

On the published cascade of diffusively coupled logistic maps, `excess3` is higher at $r = 3.85$ than at $r = 3.30$, and its grid maximum lies near the accumulation point rather than in developed chaos, at about 2.38 [Padilla-Villanueva, 2026b]. The manuscript records that location as an observation. The geometric account attached to it — a high-period orbit concentrated on a few joint symbols, still describable by pairs — is stated there as a hypothesis.

This paper tests that hypothesis. The test uses the published generator, the published seeds, and the published hybrid, on a grid refined through the accumulation band. Three comparisons are fixed in advance. The first splits the hybrid into Syn and Surp. The second asks whether the pairwise maximum-entropy law already produces the score. The third asks whether the peak survives when the window is lengthened, and whether a period-3 window repeats it. The residual Res_{pair} is reported beside the proxy and is not replaced by it.

2 Estimators

Let $S_t \in \{0, \dots, 5\}^4$ be the joint Bandt–Pompe symbol of order $m = 3$ and delay 1 at time t [Bandt and Pompe, 2002]. In a window of length w , write p for the empirical law of the joint symbol and p_i for its marginals. Total correlation, in bits, is

$$\text{TC} = \sum_{i=1}^4 H(p_i) - H(p). \quad (1)$$

The pairwise summary used by the proxy is $(4 - 1)$ times the mean pairwise mutual information. Synergy in this construction is the nonnegative excess of total correlation over that summary,

$$\text{Syn} = [\text{TC} - (d - 1) \bar{I}_{\text{pair}}]_+, \quad d = 4. \quad (2)$$

Surprise retains the positive part of the log-ratio against independence,

$$\text{Surp} = \sum_s p(s) [\log_2 p(s) - \log_2 p_{\text{ind}}(s)]_+, \quad (3)$$

with $p_{\text{ind}}(s) = \prod_i p_i(s_i)$ and the marginal probability floored at 10^{-9} in the published implementation. The hybrid is

$$\text{excess3} = 0.6 \text{Syn} + 0.4 \text{Surp}. \quad (4)$$

These names are local to the proxy. They are not the synergy and redundancy atoms of a partial information decomposition [Williams and Beer, 2010].

The strong residual is a different functional. Let $P^{(2)}$ be the maximum-entropy law with the same pairwise margins as p , obtained by iterative proportional fitting. Then

$$\text{Res}_{\text{pair}} = D_{\text{KL}}(p \parallel P^{(2)}), \quad (5)$$

in bits. A positive value of excess3 does not imply a positive value of Res_{pair} . Surprise is large for a joint that is far from independence, including a joint that pairs already explain.

3 Protocol

The system is the diffusively coupled logistic family used for the published cascade [Kaneko, 1992, Padilla-Villanueva, 2026b]. Four coordinates obey

$$\begin{aligned} f_i(t) &= r x_i(t)(1 - x_i(t)), \\ x_i(t+1) &= (1 - \varepsilon) f_i(t) + \varepsilon \bar{f}_{-i}(t) + \xi_i(t), \end{aligned} \quad (6)$$

with the mean $\bar{f}_{-i}$ taken over the other three logistic images, $\varepsilon = 0.05$, and $\xi_i(t)$ independent Gaussian noise of standard deviation 0.003. Each coordinate is then clipped to $[0, 1]$. The initial condition is uniform on $[0.2, 0.8]$. Four hundred steps are discarded and 2200 steps are retained. The seed of realization k at parameter r is

$$10000 + \lfloor 10000 r \rfloor + 17k, \quad k = 0, \dots, 5. \quad (7)$$

After the embedding, each series contains 2198 joint symbols.

The parameter grid is the 32-point cascade grid, union a uniform grid of 19 values on $[3.545, 3.590]$, restricted to $[3.05, 3.98]$. Forty-nine distinct values remain. The conventional accumulation value of the logistic family is $r_\infty = 3.5699456$ [Feigenbaum, 1978, 1983]. The nearest grid point is $r = 3.569946$. The period-3 window is represented by $r = 3.828427$.

For windows of length 13 the hybrid is the mean over every consecutive window, which is the published abundance. For every other length in $\{6, 8, 10, 13, 20, 26, 40, 52, 78, 104, 156, 208, 312\}$ the same mean is taken on about 160 evenly spaced windows, the stride being the smallest integer that yields at most that count. The full-block scores use all 2198 symbols as one window. Means and standard errors below are taken across the six seeds. The standard error uses the sample standard deviation with divisor 5.

Two Monte Carlo controls use the invariant joint of each series. Eighty sequences of length 13 are drawn independently from p , and eighty from $P^{(2)}$, with the random generator seeded by the series seed plus 901. A local control repeats the comparison inside 36 evenly spaced windows of length 13, with four draws from that window's own p and four from its pairwise projection. The film strip in Figure 2 is one realization per station: the seed whose mean number of distinct 13-symbol tuples is closest to the six-seed mean, and, inside that series, a run of 16 symbols whose support is closest to the series mean.

The window-13 means at $r = 3.30$ and $r = 3.85$ agree with the cascade archive to the stored digits, 1.757509 and 1.863299. At every station the stored means satisfy $0.6 \text{ Syn} + 0.4 \text{ Surp} = \text{excess3}$ to the precision of the archive.

4 Results

4.1 The maximum is a band, and Surp carries it

Figure 1 shows the three scores along the grid. Table 1 lists the stations used below. The largest six-seed mean of excess3 is 2.378 ± 0.002 at $r = 3.5675$. The neighboring station $r = 3.566168$, the location cited for the cascade maximum, has mean 2.378 ± 0.003 . At the nearest grid point to r_∞ the mean is 2.343 ± 0.032 . Across the 33 stations in $[3.544, 3.595]$, 29 means exceed 2.30. The lowest mean in that band is 2.204 and the highest is 2.378.

Leave-one-seed maxima of the mean fall at $r = 3.566168$ once, at $r = 3.5675$ four times, and at $r = 3.575$ once. All six lie inside the same band. The maximum is the top of a plateau.

At $r = 3.5675$ the two pieces of the hybrid are $\text{Syn} = 0.0065 \pm 0.0004$ and $\text{Surp} = 5.936 \pm 0.004$. Their weighted contributions are 0.0039 and 2.374 bits. The elevation above developed chaos at $r = 3.85$, where $\text{excess3} = 1.863 \pm 0.004$, is 0.515 bits. The Surp term contributes +0.566 bits to that difference and the Syn term contributes -0.051 bits. Along the 49 grid means, the correlation of excess3 with Surp is 0.999, with Syn it is -0.778 , and with the number of distinct tuples it is -0.975 . The largest Syn on the grid is 0.118 at $r = 3.95$.

4.2 The joint support collapses

Figure 3 places the support next to the block residual. At the grid maximum a window of 13 symbols contains 4.11 ± 0.02 distinct joint tuples, visited 3.20 times on average, with modal mass 0.304. At $r = 3.30$ the same window contains 11.39 ± 0.05 tuples. At $r = 3.85$ it contains 12.62 ± 0.02 , visited 1.04 times, with modal mass 0.098.

Figure 2 shows sixteen consecutive symbols from the representative realization at each of those three stations. At $r = 3.5675$ the segment is the cycle of four joint symbols 1, 2, 3, 4, repeated four times. The integers are the order of first appearance in the segment. They label distinct 4-tuples of permutation indices; they are not comparable across panels, and they are not themselves the Bandt–Pompe ranks. At $r = 3.30$ the segment uses eleven joint symbols. At $r = 3.85$ it uses thirteen.

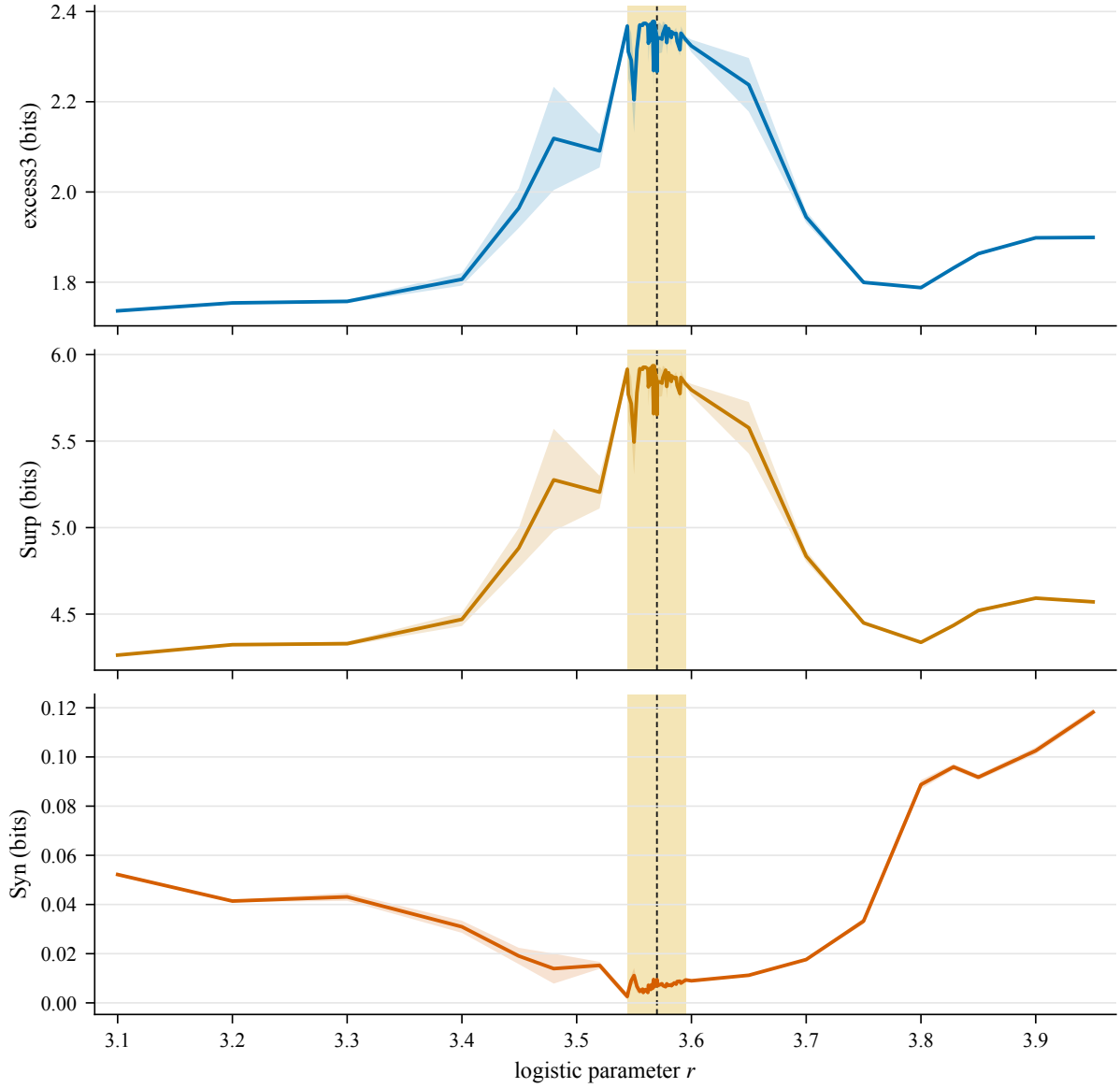


Figure 1: Window-13 scores along the logistic parameter. Curves are six-seed means and shading is the standard error. The band marks $r \in [3.544, 3.595]$. The dashed line is $r_\infty = 3.5699456$. Surp follows excess3. Syn is near zero inside the band and is largest at the right edge of the grid.

Alt text. Three stacked curves against the logistic parameter. Excess3 and surprise both rise to a plateau near 3.57 and then fall. Synergy is near zero on that plateau and rises only after 3.75.

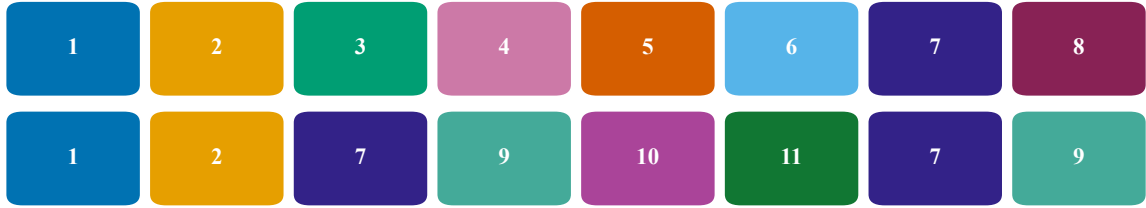
Table 1: Six-seed means at nine stations. The standard error of the mean is in parentheses. Tuples, visits, and modal mass refer to windows of length 13. Res_{pair} is the Kullback–Leibler residual of the full symbol block.

r	regime	excess3	Syn	Surp	tuples	visits	modal	Res_{pair}
3.30000	period 2	1.758 (0.004)	0.0431 (0.0016)	4.329 (0.012)	11.39 (0.05)	1.15	0.147	0.0024 (0.0004)
3.44949	period 4	1.964 (0.044)	0.0191 (0.0033)	4.882 (0.115)	8.91 (0.55)	1.64	0.190	0.0018 (0.0006)
3.54409	period 8	2.368 (0.008)	0.0026 (0.0014)	5.915 (0.021)	4.20 (0.09)	3.16	0.302	0.0005 (0.0002)
3.56617	near r_{∞}	2.378 (0.003)	0.0059 (0.0002)	5.935 (0.007)	4.12 (0.03)	3.20	0.304	0.0005 (0.0003)
3.56750	grid maximum	2.378 (0.002)	0.0065 (0.0004)	5.936 (0.004)	4.11 (0.02)	3.20	0.304	0.0003 (0.0001)
3.56995	r_{∞}	2.343 (0.032)	0.0074 (0.0009)	5.845 (0.083)	4.51 (0.36)	3.04	0.293	0.0031 (0.0028)
3.82843	period-3 window	1.832 (0.003)	0.0960 (0.0012)	4.435 (0.008)	12.56 (0.02)	1.04	0.101	0.1938 (0.0052)
3.85000	developed chaos	1.863 (0.004)	0.0917 (0.0011)	4.521 (0.010)	12.62 (0.02)	1.04	0.098	0.1921 (0.0016)
3.95000	largest Syn	1.899 (0.003)	0.1182 (0.0014)	4.571 (0.008)	12.79 (0.02)	1.02	0.090	0.1816 (0.0025)

Alt text. A table of nine logistic parameter values. Excess3 and surprise are highest near $r = 3.5675$, where about four joint tuples appear. The full-block residual is near zero there and near 0.19 in the period-3 window and in developed chaos.

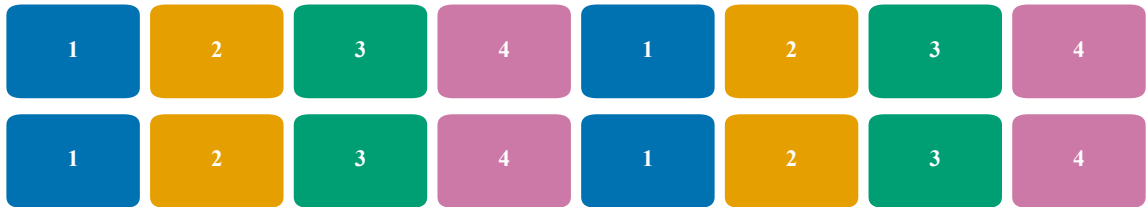
The block residual does not follow the proxy. At $r = 3.5675$ it is 0.0003 ± 0.0001 . At $r = 3.85$ it is 0.192 ± 0.002 , and its grid maximum is 0.198 ± 0.003 at $r = 3.80$. The full-block hybrid itself remains high through the accumulation band. Its grid maximum is 2.298 at $r = 3.5575$. At $r = 3.85$ the full-block hybrid has fallen to 0.174 ± 0.003 .

(a) Period 2 $r = 3.30$



11 distinct joint symbols in these 16 steps

(b) Accumulation $r = 3.5675$



4 distinct joint symbols in these 16 steps

(c) Developed chaos $r = 3.85$



13 distinct joint symbols in these 16 steps

Figure 2: Sixteen consecutive joint symbols, read left to right and then onto the second row. The displayed integer is the order of first appearance within the segment. The segment is chosen by the rule in Section 3. (a) Period 2 uses eleven joint symbols. (b) At the grid maximum the same four joint symbols repeat. (c) Developed chaos uses thirteen joint symbols in the same length.

Alt text. Three pairs of colored rows. The middle pair repeats the colors and numbers 1, 2, 3, 4. The top and bottom pairs introduce many more colors.

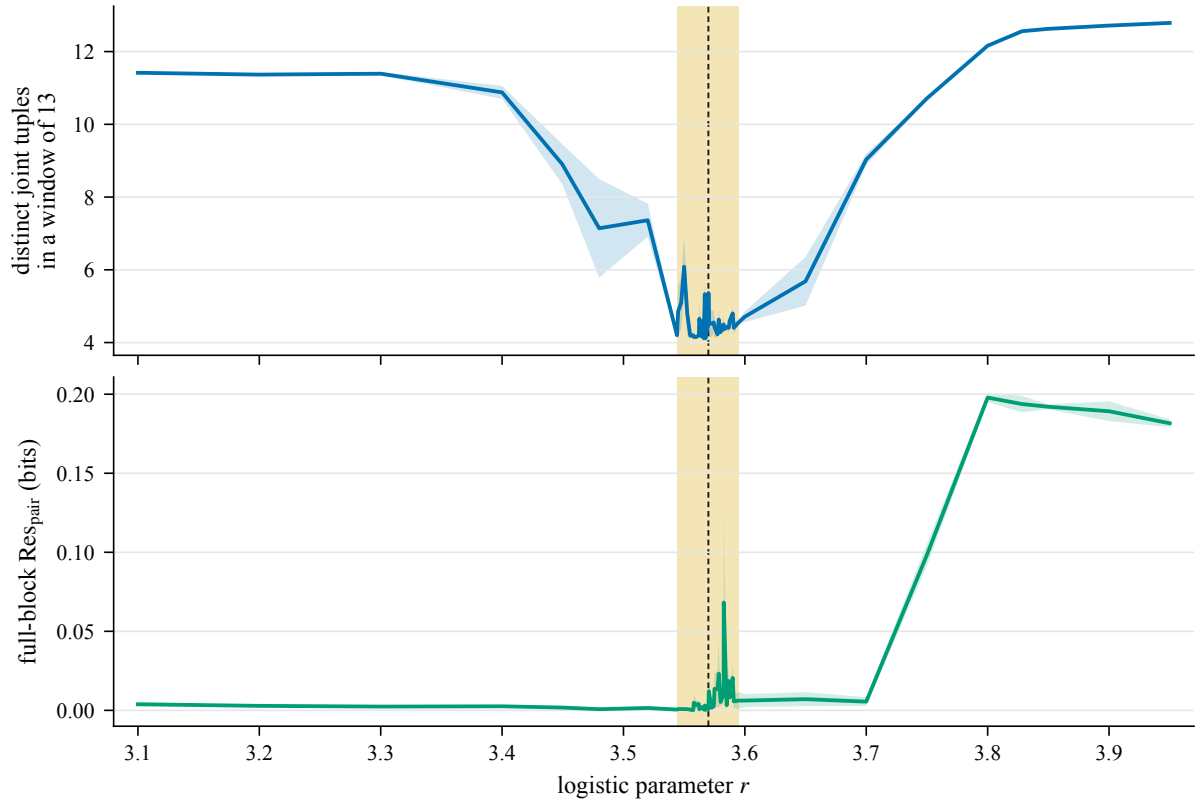


Figure 3: Support and block residual along r . Shading, band, and dashed line are as in Figure 1. The number of distinct joint tuples in a window of 13 is smallest inside the band. The full-block residual stays near zero there, apart from isolated grid points whose standard error is comparable to the mean: at $r = 3.582612$ the residual mean is 0.068 with standard error 0.059. The residual then rises and reaches 0.198 ± 0.003 at $r = 3.80$.

Alt text. Two stacked curves. The count of distinct joint tuples falls to about four inside the yellow band and then rises past twelve. The full-block residual stays near zero through that band and rises to about 0.20 after $r = 3.75$.

4.3 Pairs reproduce the stationary score

Table 2 compares the sliding window with the two independent-draw experiments. At the grid maximum, draws from the empirical joint p score 2.073 ± 0.067 , and draws from $P^{(2)}$ score 2.088 ± 0.062 . The ratio of the pairwise score to the sliding score is 0.878. At $r = 3.85$ the same ratio is 0.910. The pairwise projection therefore accounts for the excess3 of the invariant joint. What it does not account for is the extra concentration of consecutive symbols: the sliding mean exceeds the independent draws from p by 0.305 ± 0.067 bits at the maximum, about one eighth of the sliding score. Those sliding windows occupy 4.11 tuples, while the independent draws from the same p occupy 5.54.

Inside single windows the comparison is tighter. At the grid maximum the gap between a window and draws from its own pairwise projection is 0.164 ± 0.002 , and the gap against draws from that window's own p is 0.146 ± 0.006 . The two gaps agree at the scale of the Monte Carlo error. At $r = 3.85$ the local pairwise gap is -0.035 ± 0.008 . The short window does not contain a residual beyond pairs above the sampling fluctuation of a 13-symbol table.

4.4 The peak survives a longer window

Figure 4 extends the hybrid from length 6 to length 312. At $r = 3.5675$ the mean is 2.326 at length 6, 2.378 at length 13, and 2.312 ± 0.027 at length 312. At that last length the accumulation series still occupies only 6.5 ± 0.2 joint symbols, with 55.8 visits per occupied tuple. At $r = 3.85$ the mean falls from 2.078 at length 6 through 1.863 at length 13 and 0.912 at length 104 to 0.514 ± 0.006 at length 312, where the window occupies 199.5 ± 1.2 symbols and only 1.57 visits per tuple. Lengthening the window enlarges the contrast. It does not create the peak.

The period-3 station $r = 3.828427$ follows the chaotic curve. Its window-13 hybrid is 1.832 ± 0.003 , its support is 12.56 ± 0.02 tuples, its synergy is 0.096 ± 0.001 , and its block residual is 0.194 ± 0.005 . A periodic window is not, by itself, the accumulation peak.

Period 2 behaves in a third way. By length 312 its joint support has saturated at 32 symbols and excess3 has leveled at 1.224, between the accumulation plateau and the chaotic decline. The period-8 station $r = 3.54409$ is already inside the high band: excess3 = 2.368 ± 0.008 and the short window holds 4.20 ± 0.09 tuples.

Table 2: Grid maximum against developed chaos. Entries are six-seed means $\pm$ the standard error. Independent draws are eighty sequences of length 13. Local gaps use 36 windows and four draws each.

quantity	$r = 3.5675$	$r = 3.85$
excess3, $w = 13$	2.378 ± 0.002	1.863 ± 0.004
Syn	0.0065 ± 0.0004	0.0917 ± 0.0011
Surp	5.936 ± 0.004	4.521 ± 0.010
distinct tuples, $w = 13$	4.11 ± 0.02	12.62 ± 0.02
visits per tuple	3.20 ± 0.01	1.04 ± 0.00
modal mass	0.3044 ± 0.0005	0.0981 ± 0.0008
iid draws from P	2.073 ± 0.067	1.691 ± 0.012
iid draws from $P^{(2)}$	2.088 ± 0.062	1.695 ± 0.008
sliding minus iid P	0.305 ± 0.067	0.172 ± 0.012
local gap vs. $P^{(2)}$	0.164 ± 0.002	-0.035 ± 0.008
local gap vs. own P	0.146 ± 0.006	-0.039 ± 0.007
full-block excess3	2.136 ± 0.107	0.174 ± 0.003
full-block Res _{pair}	0.0003 ± 0.0001	0.1921 ± 0.0016
excess3, $w = 312$	2.312 ± 0.027	0.514 ± 0.006
distinct tuples, $w = 312$	6.5 ± 0.2	199.5 ± 1.2

Alt text. A two-column comparison of the accumulation maximum and developed chaos. Surprise, repeated tuples, and the sliding-window score are higher at the maximum. Synergy and the full-block residual are higher in chaos. Independent draws from the joint and from its pairwise projection agree with each other in both columns.

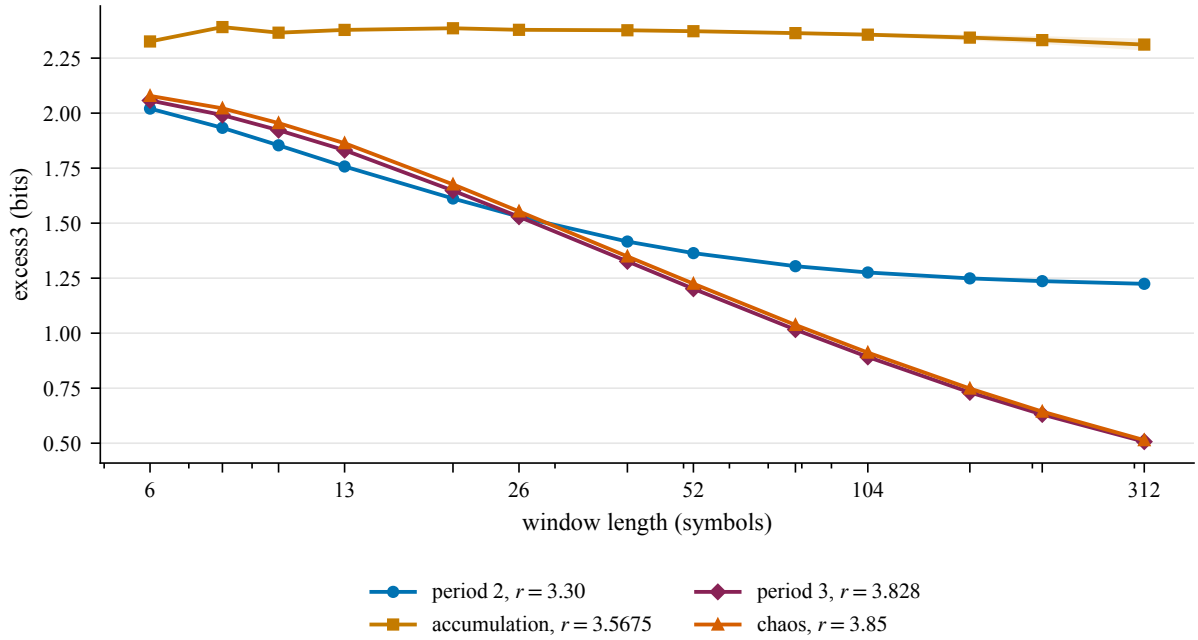


Figure 4: The hybrid against window length at four stations. Points are six-seed means. Lengths above 13 use about 160 evenly spaced windows. The accumulation curve stays near 2.3. The period-3 and chaotic curves fall together toward 0.5. Period 2 falls and then levels off.

Alt text. A log-scaled plot of excess3 against window length. The accumulation curve is nearly flat near 2.3. Period 2 declines and flattens near 1.2. Period 3 and developed chaos decline together to about 0.5 at window length 312.

5 Discussion

The measurement closes the geometric hypothesis on this ensemble. Near accumulation the orbit of the joint symbol lives on about four tuples. Those tuples are far from an independent product, so Surp is large, and the declared hybrid follows Surp . The same joint is still close to its pairwise maximum-entropy projection, so Res_{pair} on the long block is indistinguishable from zero at the reported precision. Developed chaos widens the support, lowers Surp , and is the region in which the block residual becomes nonzero. Synergy, as defined by equation (2), is the term that rises in that later region, and on this grid it never sets the location of the hybrid maximum.

The two quantities therefore mark different stations of the same cascade. A reading that takes the peak of excess3 as the place where ordinal structure beyond pairs is strongest points to the accumulation band. A reading that takes $\text{Res}_{\text{pair}} > 0$ as that place points to developed chaos, after the proxy has fallen. The foundations manuscript already separates the proxy from the residual and already distinguishes a common-driver geometry, in which surprise is high and the pairwise residual is not, from a higher-order joint [Padilla-Villanueva, 2026b]. The accumulation band of this coupled logistic system is the first of those geometries. The chaotic stations are where the second geometry begins to register in the block residual.

Two smaller facts keep the statement from being sharper than the ensemble. First, consecutive windows are more concentrated than independent draws from the invariant joint, by about 0.30 bits at the maximum. That is a real temporal contribution. It is smaller than the surprise of the joint, and the local 13-symbol comparison does not resolve a pairwise gap beyond the resampling error of the window. Second, one grid point inside the band, $r = 3.582612$, has a block-residual mean of 0.068 with standard error 0.059. Isolated seeds can move a single station. They do not move the six-seed maximum of the residual, which remains at $r = 3.80$.

The scope is the generator in equation (6): four coordinates, coupling 0.05, noise level 0.003, embedding order 3, and the six seeds in equation (7). The weights in equation (4) are the published weights. Because Syn is three orders of magnitude smaller than Surp at the maximum, a convex combination that still gives Surp positive weight keeps the maximum on the surprise curve; the paper does not refit the weights. The result does not derive the operational concordance bands used elsewhere in the framework, and it does not transfer the peak, or the delayed rise of Res_{pair} , to other couplings, other dimensions, or series measured in the field. Early-warning comparisons with variance and lag-1 autocorrelation remain those of the cascade study.

6 Conclusion

On this coupled logistic cascade, excess3 peaks near accumulation because the joint ordinal symbol repeats a handful of tuples and those tuples are far from independence. The part of the construction that is meant to see beyond pairs is quiet there. It becomes visible later, in developed chaos, after the proxy has come down. The period-3 window follows that later regime. A longer window preserves the peak and widens its contrast with chaos.

Author declarations

Conflict of interest. The author has no conflict of interest to disclose.

Ethics approval. Ethics approval is not required. The study uses simulated series only.

Author contribution. Johel Padilla-Villanueva: conceptualization, methodology, software, formal analysis, and writing.

Data availability

This article is archived at <https://doi.org/10.5281/zenodo.22945800>. The generator, the seeds, and the estimators are those of the cascade in Padilla-Villanueva [2026b] and of nested-recd 0.2.2 [Padilla-Villanueva, 2026c]. The decomposition archived for this article is the run of 24 September 2026. The files `peak_flat_20260924_145527.csv`, `peak_agg_20260924_145527.csv`, and `peak_results_20260924_145527.json` accompany this deposit and the repository <https://github.com/johelpadilla/excess3> under `accumulation/data/`. They were produced by `scripts/peak_rinf_decompose.py`. The figures and the numeric tables were drawn from that archive by `make_figures.py`. The cascade and diagnostic scripts of the foundations study are `cascade_r_sweep.py` and `res_pair_diagnostics.py`.

References

- Christoph Bandt and Bernd Pompe. Permutation entropy: a natural complexity measure for time series. *Physical Review Letters*, 88(17):174102, 2002. doi: 10.1103/PhysRevLett.88.174102.
- Mitchell J. Feigenbaum. Quantitative universality for a class of nonlinear transformations. *Journal of Statistical Physics*, 19(1):25–52, 1978. doi: 10.1007/BF01020332.
- Mitchell J. Feigenbaum. Universal behavior in nonlinear systems. *Physica D*, 7(1–3):16–39, 1983. doi: 10.1016/0167-2789(83)90112-4.
- Kunihiko Kaneko. Overview of coupled map lattices. *Chaos*, 2(3):279–282, 1992. doi: 10.1063/1.165869.
- Johel Padilla-Villanueva. excess3: A pre-specified continuous proxy for order-3 synergistic surplus. Zenodo, 2026a. URL <https://doi.org/10.5281/zenodo.21385937>.
- Johel Padilla-Villanueva. Systemic Tau and hierarchical ordinal conjunctions: A relational theory of critical transitions. Zenodo, 2026b. URL <https://doi.org/10.5281/zenodo.21753560>.
- Johel Padilla-Villanueva. nested-recd 0.2.2: Nested ordinal RECD, excess3 proxy, and Res_pair, 2026c. URL <https://doi.org/10.5281/zenodo.21937204>.
- Marten Scheffer, Jordi Bascompte, William A. Brock, Victor Brovkin, Stephen R. Carpenter, Vasilis Dakos, Hermann Held, Egbert H. van Nes, Max Rietkerk, and George Sugihara. Early-warning signals for critical transitions. *Nature*, 461:53–59, 2009. doi: 10.1038/nature08227.
- Paul L. Williams and Randall D. Beer. Nonnegative decomposition of multivariate information. *arXiv preprint arXiv:1004.2515*, 2010.